

Day to Day Stock Price Predictor

Next-day directional prediction for MSFT powered by machine learning.

TOMORROW · MON MAY 19



MODEL CONFIDENCE

67%

ENSEMBLE BREAKDOWN

Random Forest

UP · 71%

XGBoost

UP · 65%

LSTM

UP · 58%

Stacking Meta

UP · 67%

Download report (CSV)

View prediction details →

LAST CLOSE PRICE

\$458.21

Fri May 16, 2026 · +1.42%

PREDICTION DATE

Mon May 19, 2026

Next trading session

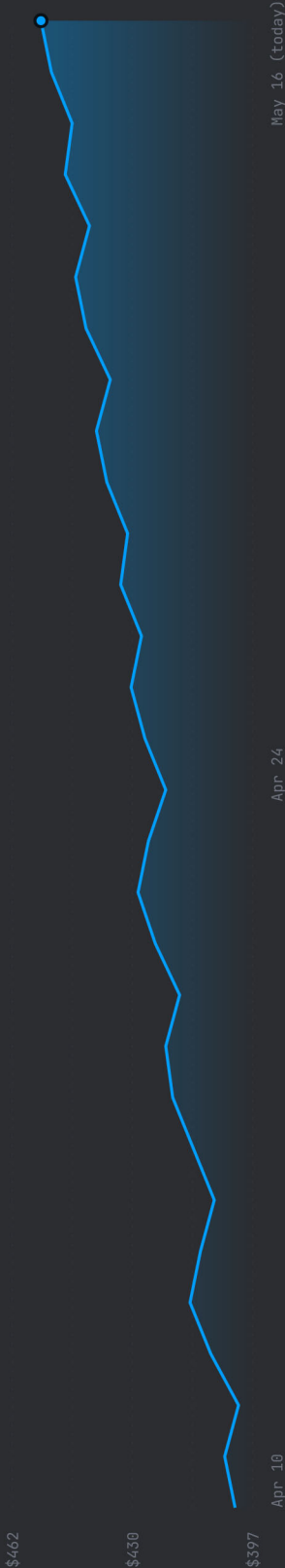
MODEL USED

Stacking Ensemble

RF + XGB + LSTM → Logistic meta

MSFT · Last 30 trading days

Close price · USD · with next-day prediction annotated



⚠️ **DISCLAIMER** This tool is for educational purposes only and does not constitute financial advice.

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- Model Performance
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Historical Price Analysis

Explore MSFT closing price from 2015 to 2025 and overlay macro indicators.

START DATE

2015-01-02

END DATE

2025-12-31

Reset

Apply

MSFT Closing Price - 2015-2025

Toggle overlays to compare with macro indicators

MSFT

☒

Gold (GLD)

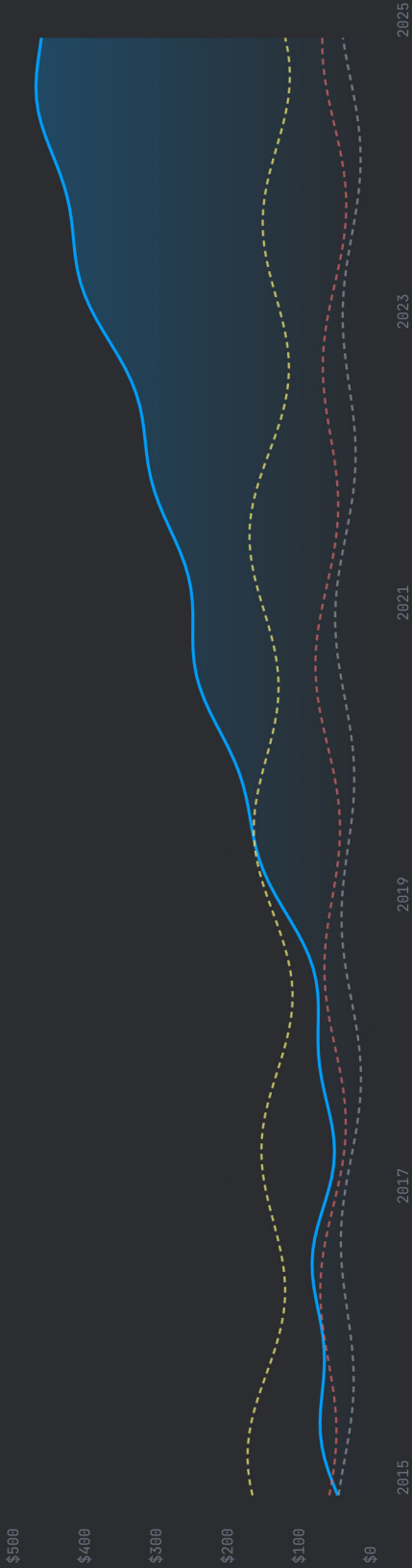
☒

Crude (OIL=F)

☒

VIX

☐



Daily Return %

positive negative

+3%

0%

-3%

Last 60 sessions



Summary Statistics

Mean Return	+0.082%
Volatility (σ)	1.74%
Max Drawdown	-34.6%
Best Day	+14.22% · Mar 13 2020
Worst Day	-14.74% · Mar 16 2020
Sharpe (3y)	1.12

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Model Performance

Stacking ensemble · evaluated on test set Feb 2023 – Feb 2025 (380 trading days)

ACCURACY

0.79

301 / 380 correct

PRECISION

0.81

True UP / predicted UP

MODEL

Stacking Ensemble

TEST WINDOW

Feb 23 – Feb 25

RECALL

0.80

True UP / actual UP

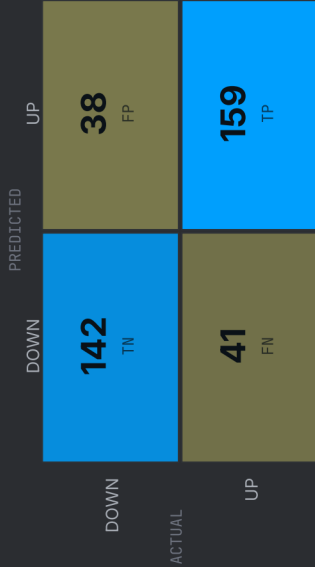
F1 SCORE

0.79

harmonic mean

Confusion Matrix

Counts on the held-out test set



TN True
Negative

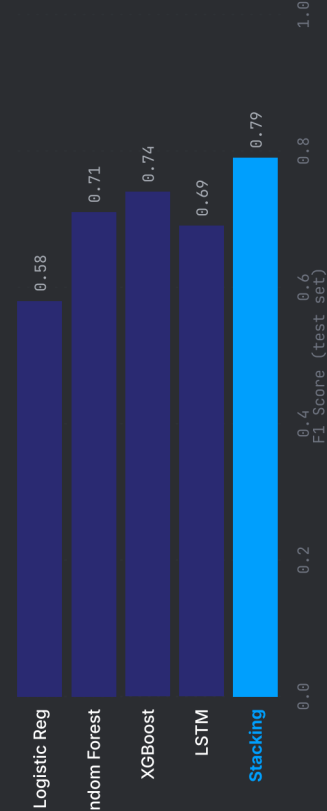
FP False
Positive

FN False
Negative

TP True
Positive

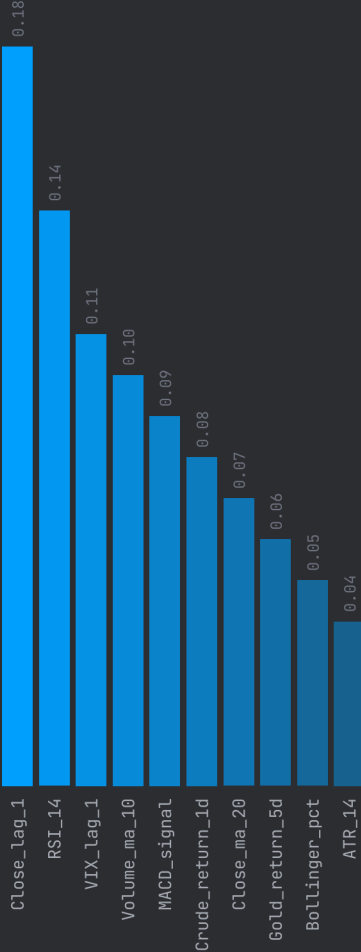
Model Comparison · F1 Score

Stacking ensemble (highlighted) outperforms each base model individually.



Feature Importance

Top 10 features ranked by XGBoost gain



Toggle source: XGBoost

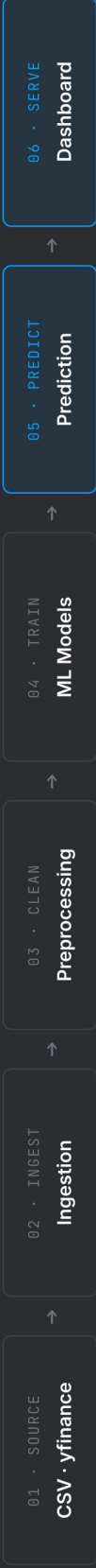
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About & Methodology

How the prediction is made, what data feeds it, and who built it.

The **Day to Day Stock Price Predictor** estimates whether Microsoft's (MSFT) closing price will move **up** or **down** on the next trading day. It combines three machine-learning models — a Random Forest, an XGBoost classifier, and an LSTM neural network — into a **stacking ensemble** that learns when each underlying model is most trustworthy. Inputs include MSFT daily prices and three macro signals (Gold, Crude Oil, and the VIX volatility index). The dashboard is built for retail investors who want a simple directional signal without wading through code or formulas.

Pipeline



Daily cron · 09:30 ET · before market open

Data Sources

ASSET	SOURCE	FORMAT	UPDATE FREQUENCY
MSFT	Yahoo Finance (yfinance API)	CSV / DataFrame	Daily · post-close 16:00 ET
Gold (GLD)	Yahoo Finance (yfinance API)	CSV / DataFrame	Daily · post-close 16:00 ET
Crude Oil (CL=F)	Yahoo Finance (yfinance API)	CSV / DataFrame	Daily · post-close 17:00 ET
VIX (^VIX)	CBOE via yfinance	CSV / DataFrame	Daily · post-close 16:15 ET

BUILT BY

Group 15 · Block D 2025–2026

Breda University of Applied Sciences · Data Science & AI

